

SDLC Magento 2 Connector

for Odoo 18

User Guide

Version 1.0

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Module: `sdlc_magento_connector`

A comprehensive guide to installing, configuring, and using the SDLC Magento 2 Connector module for bidirectional synchronization between Odoo 18 and Magento 2 stores.

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1. Overview & Key Features

The **SDLC Magento 2 Connector** provides deep, bidirectional integration between Odoo 18 and Magento 2 e-commerce stores. It synchronizes products, customers, orders, categories, invoices, shipments, and credit memos in real time.

Key Features

- **Multi-Instance Support** - Connect multiple Magento 2 stores simultaneously
- **Bidirectional Field Mapping** - Map any Odoo field to any Magento API field including custom attributes
- **Real-time Dashboard** - KPIs, revenue charts, order status, top categories, top countries, sync health
- **Auto Sync Cron** - Schedule automatic sync (hourly/daily/weekly/monthly) per instance
- **Complete Order Lifecycle** - Orders, Invoices, Shipments, Credit Memos fully synced
- **Product Sync** - SKU-based mapping, images, stock, descriptions, categories, attribute sets
- **Customer Sync** - Full profile with addresses, gender, DOB, tax/VAT, groups
- **Inventory Tracking** - Real-time stock status sync between Magento and Odoo
- **Sync Reports** - Detailed audit trail of every sync operation
- **Guest Cart API** - Orders pushed via Magento cart API for proper workflow
- **Auto-Sync on Save** - Products, customers, orders auto-push on create/update

2. Installation & Requirements

Prerequisites

- Odoo 18 Community or Enterprise Edition
- Python 3.10+ with **requests** library
- Magento 2.x with REST API enabled
- Magento Integration Access Token (Admin > System > Integrations)
- Odoo modules: base, product, sale, sale_stock, account, stock

Installation Steps

Step 1: Copy **sdic_magento_connector** folder into your Odoo custom addons directory.

Step 2: Restart Odoo with **-u sdic_magento_connector** or install from Apps menu.

Step 3: Search for "Magento" in Apps, click **Install**.

Step 4: The **Magento** menu appears in the top navigation bar.

Magento API Token Setup

1. In Magento Admin, go to **System > Integrations**.
2. Click **Add New Integration**.
3. Enter a name (e.g., "Odoo Connector").
4. Under **API** tab, select **All** resources.
5. Save and **Activate** the integration.
6. Copy the **Access Token** for use in Odoo.

3. Instance Configuration

Navigate to **Magento > Instances** to manage store connections.

Creating a New Instance

Field	Description	Example
Instance Name	Friendly name for this store	EU Magento Store
Base URL	Magento store URL	https://magento.example.com
Access Token	API integration token	abc123xyz...
Verify SSL	Enable SSL certificate verification	Checked
Website ID	Magento website ID (usually 1)	1
Store ID	Magento store ID	1
Store Code	Store view code	default
Root Category ID	Root category for products	2
Attribute Set Skeleton	Template for new attribute sets	4
Default Payment Method	Payment method code	checkmo
Default Shipping Method	Shipping method code	flatrate_flatrate
Default Shipping Amount	Shipping cost	5.00
Customer Default Password	For customer creation	Change@123

Testing the Connection

Click **Test Connection** in the form header. A success notification confirms the API is reachable. If it fails, check the URL and token.

Sync Buttons

Button	Action
Sync Products	Pull all products from Magento into Odoo
Sync Categories	Pull category tree from Magento
Sync Customers	Pull all customers with addresses
Sync Orders	Pull orders with invoices, shipments, credit memos
Sync Attribute Sets	Pull product attribute set definitions

4. Dashboard

Navigate to **Magento > Dashboard** for a real-time analytics overview.

Dashboard Sections

Header Controls

- **Instance Selector** - Filter by specific Magento store or view all
- **Date Range** - Last 8 / 14 / 30 / 60 days
- **Refresh Button** - Reload all data
- **Auto-refresh** - Dashboard refreshes every 30 seconds

KPI Cards

Card	Description	Color
Total Amount	Sum of order totals + pending amount	Orange gradient
Total Orders	Order count with change indicator	Purple gradient
Delivered Orders	Orders marked as delivered	Green gradient
Not Delivered	Orders pending delivery	Pink gradient
Total Customers	Synced customer count + change	Blue gradient
Total Products	Synced product count + change	Teal gradient

Charts & Analytics

- **Revenue Analytics** - Line chart: daily revenue + order count over selected period
- **Order Status Mix** - Stacked progress bars showing status distribution
- **Top Categories** - Donut chart with category revenue breakdown
- **Top Countries** - Horizontal bars showing order distribution by country

Sync Health

- Manual Runs / Cron Runs counters
- Success Records / Error Records counters
- Last sync timestamp, Avg Order Value, Delivered Rate

5. Product Management

Navigate to **Magento > Magento Data > Products**.

Product Mapping Model

Each Magento product has a **magento.product.map** record that links to both the Magento product and an Odoo **product.product** record.

Field	Description
Instance	Which Magento store
Magento ID	Magento product ID (auto-filled on sync)
SKU	Product SKU - used for matching
Name	Product name
Price	Product price
Attribute Set	Magento attribute set
Status	Enabled / Disabled
Visibility	Not Visible / Catalog / Search / Both
Type	Simple / Virtual / Downloadable
Weight	Product weight
Quantity	Stock quantity
Stock Status	In Stock / Out of Stock
Categories	Linked Magento categories
Description / Short Description	Product descriptions
Image	Product image (synced from Magento)
Odoo Product	Linked Odoo product.product record

Product Actions

- **Create in Magento** - Push new product to Magento (requires SKU)
- **Update in Magento** - Push changes to existing Magento product
- **Pull from Magento** - Refresh data from Magento
- **Update Mapped Fields** - Push only fields configured in field mapping
- **Refresh Inventory** - Sync stock quantities

Auto-Sync: When you edit a product mapping and save, changes are automatically pushed to Magento (unless skip_magento_sync context is set).

6. Customer Management

Navigate to **Magento > Magento Data > Customers**.

Field	Description
Instance	Magento store instance
Magento ID	Magento customer ID
Email	Required - used for matching
Name Fields	Prefix, First, Middle, Last, Suffix
Phone	Phone number
Date of Birth	Customer birthday
Gender	Male / Female / Not Specified
Tax/VAT	Tax identification number
Group/Website/Store ID	Magento assignment IDs
Address Fields	Street, City, Region, Postcode, Country
Default Billing/Shipping	Address type flags

Customer Actions

- **Create in Magento** - Creates customer with configured default password
- **Update in Magento** - Updates profile (uses fallback minimal payload if needed)
- **Pull from Magento** - Refresh from Magento API
- **Auto-Sync**: Customers auto-push on create and update

Warning: Customer creation requires a default password configured on the instance. Password must meet Magento's minimum length requirement.

7. Order Management

Navigate to **Magento > Sales > Orders**.

Magento Order Model

Field	Description
Instance	Magento store
Magento ID / Increment ID	Magento order identifiers
Status	Magento order status (pending, processing, complete, etc.)
Customer	Linked Odoo partner
Grand Total / Line Total	Order amounts
Currency	Order currency code
Order Lines	Items with SKU, quantity, price, subtotal
Created At / Updated At	Magento timestamps

Order Sync Workflow

Pull from Magento (Sync Orders):

1. Fetch all orders from Magento API
2. For each order: resolve/create customer (res.partner)
3. Create magento.order + line items
4. Create linked sale.order in Odoo Sales module
5. Sync invoices, shipments, and credit memos
6. Register payments if auto_register_magento_payment is enabled

Push to Magento (Create Order):

1. Create a guest cart via Magento Cart API
2. Add items to cart (resolves SKUs)
3. Set shipping address + shipping method
4. Set billing address + payment method
5. Place order - Magento returns order ID

8. Invoice, Shipment & Credit Memos

The connector extends Odoo's Sale Order, Stock Picking, and Account Move models.

Sale Order Enhancements

- Magento tab showing order ID, status, timestamps
- Invoice State tracking (Not Invoiced / Partial / Invoiced)
- Payment State tracking (Not Paid / Partial / Paid / Refunded)
- Total Invoiced / Total Paid monetary fields
- Create/Update/Pull buttons for Magento sync

Invoice Sync (Account Move)

- When an invoice is **posted** in Odoo, it auto-creates an invoice in Magento
- Magento invoice ID stored on the Odoo invoice
- When invoice is **paid**, Magento order status updated (if configured)
- Credit memos synced from Magento with refund amounts

Shipment Sync (Stock Picking)

- When delivery is **validated** (Done), shipment auto-pushes to Magento
- **Push Shipment to Magento** button for manual sync
- Magento shipment ID stored on the delivery order
- Tracks: ship-to name, customer email, phone, total qty, dates

Credit Memo Sync

- Credit memos pulled from Magento during order sync
- Tracks: credit memo ID, grand total, tax, subtotal, refund state
- Smart button on invoice to view linked credit memos

9. Field Mapping (Bidirectional)

Navigate to **Magento > Mapping** to configure custom field mappings.

Mapping Types

- **Product Mapping** - Map product.product / magento.product.map fields to Magento product API fields
- **Customer Mapping** - Map magento.customer fields to Magento customer API fields
- **Category Mapping** - Map magento.category fields to Magento category API fields

How Mapping Works

Column	Description
Mapping Type	Product / Customer / Category
Instance	Which Magento store this mapping applies to
Odoo Field	Dropdown of all stored fields on the target Odoo model
Magento Field	Dropdown of all Magento API fields (auto-fetched from Magento)
Active	Enable/disable this individual mapping
Test	Validates the mapping and shows a preview report

Loading Available Fields

Click **Load New Fields** button to fetch all available fields from your Magento instance. This queries the Magento API for product attributes, customer attributes, and category attributes, creating **magento.available.field** records.

Advanced Mapping Features

- **Nested Fields** - Support for dot-notation (e.g., extension_attributes.stock_item.qty)
- **Custom Attributes** - Maps to custom_attributes array (e.g., custom_attributes.color)
- **Address Fields** - Maps to addresses array (e.g., addresses.street)
- **Type Coercion** - Auto-converts between Odoo and Magento field types
- **Outbound Mapping** - Odoo field values injected into Magento API payload on push
- **Inbound Mapping** - Magento API response values set on Odoo fields on pull

Testing Mappings

Click the **Test** button on any mapping row. This generates a **Mapping Report** showing the field-by-field changes and the full API payload. View reports at **Magento > Reports > Mapping Report**.

10. Auto Sync Cron Jobs

Configure per-instance on the **Instance form > Auto Sync Cron** tab.

Schedule Settings

Field	Description
Auto Sync Enabled	Master toggle
Frequency	Hourly / Daily / Weekly / Monthly
Hour / Minute	Time of day to run (server time)
Sync Products	Include products in auto sync
Sync Customers	Include customers
Sync Categories	Include categories
Sync Orders	Include orders (with invoices, shipments, credit memos)

Click **Apply Auto-Sync Cron** to create/update the scheduled action. The cron generates a **Sync Report** for each run.

Built-in Crons

Cron	Interval	Action
Magento Product Sync	Every 10 minutes	Sync products for all instances
Magento Order Sync	Every 10 minutes	Sync orders for all instances

11. Inventory Management

Navigate to **Magento > Inventory**.

The inventory view shows all Magento product mappings with stock data:

Column	Description
Product	Product name
SKU	Product SKU
Quantity	Current stock quantity
Stock Status	In Stock / Out of Stock
Instance	Magento store

Use the **Refresh Inventory** button on individual products to re-sync stock from Magento.

12. Sync Reports & Logs

Navigate to **Magento > Reports > Sync Reports**.

Field	Description
Instance	Which Magento store
Synced At	Timestamp of the sync
Sync Type	All / Category / Product / Order / Customer
Mode	Manual / Cron
Source Action	Button or method that triggered the sync
Total Records	Total records processed
Success / Errors	Breakdown of results
Categories Created/Updated	Category sync details
Products Created/Updated	Product sync details
Orders Created/Updated	Order sync details
Customers Created/Updated	Customer sync details
Inventory Updated	Stock updates count
Notes	Additional sync details or errors

13. Menu Structure

Menu Item	Description
Magento	Main menu
Dashboard	Real-time analytics with charts and KPIs
Instances	Magento store connections
Magento Data	Parent menu
Products	Magento product mappings
Customers	Magento customers
Categories	Magento categories
Mapping	Parent menu
Product Mapping	Product field mappings
Customer Mapping	Customer field mappings
Category Mapping	Category field mappings
Sales	Parent menu
Orders	All Magento-linked sale orders
Delivered Orders	Completed deliveries
Not Delivered Orders	Pending deliveries
Invoices	Magento-linked invoices
Shipments	Magento-linked shipments
Credit Memos	Magento credit memos
Inventory	Stock status overview
Reports	Parent menu
Sync Reports	Sync audit trail
Mapping Report	Field mapping test results
Cancelled Orders	Cancelled order list

14. Magento API Reference

The connector uses Magento 2 REST API with Bearer token authentication.

API Endpoints Used

Endpoint	Method	Purpose
/rest/V1/products	GET	List all products
/rest/V1/products/{sku}	GET/PUT	Get/update product by SKU
/rest/V1/products	POST	Create new product
/rest/V1/products/{sku}/media	POST/PUT	Product images
/rest/V1/categories	GET/POST	List/create categories
/rest/V1/categories/{id}	GET/PUT	Get/update category
/rest/V1/customers/search	GET	Search customers
/rest/V1/customers	POST	Create customer
/rest/V1/customers/{id}	GET/PUT	Get/update customer
/rest/V1/orders	GET	List orders
/rest/V1/orders/{id}	GET	Get order
/rest/V1/order/{id}/invoice	POST	Create invoice
/rest/V1/order/{id}/ship	POST	Create shipment
/rest/V1/order/{id}/refund	POST	Create credit memo
/rest/V1/guest-carts	POST	Create guest cart
/rest/V1/guest-carts/{id}/items	POST	Add cart items
/rest/V1/eav/attribute-sets/list	GET	Attribute sets

API Fallback Logic

The connector tries multiple API prefixes in order: **/rest/all/V1** then **/rest/{store_code}/V1** then **/rest/V1**. Container-aware fallback (localhost:8080, host.docker.internal) is also supported.

15. Troubleshooting

Issue: Connection test fails

- Verify the Base URL is correct and reachable from the Odoo server
- Check the Access Token is valid and not expired
- Try with Verify SSL unchecked if using self-signed certificates
- Check firewall rules between Odoo and Magento servers

Issue: Products not syncing

- Ensure connection is established first
- Check Sync Reports for detailed error messages
- Verify API token has product resource permissions
- Check if products exist in the configured website/store

Issue: Orders fail to create in Magento

- Verify Default Payment Method code matches Magento config
- Verify Default Shipping Method code matches Magento config
- Ensure products have valid SKUs that exist in Magento
- Check customer has a valid email and address

Issue: Field mapping not applying

- Ensure mappings are Active (toggled on)
- Click "Load New Fields" first to fetch Magento fields
- Use the Test button to validate each mapping
- Check mapping type matches the entity you're syncing

Issue: Invoice/shipment not syncing to Magento

- Ensure the sale order has a Magento Order ID
- Invoice must be Posted before it syncs
- Delivery must be Done (validated) before shipment syncs
- Check if `auto_register_magento_payment` is enabled on instance

16. FAQ

Q: Can I connect multiple Magento stores?

A: Yes. Create a separate instance for each store. Products, orders, and customers are isolated per instance.

Q: Does the connector support Magento custom attributes?

A: Yes. The field mapping system supports custom_attributes via dot notation (e.g., custom_attributes.color).

Q: How are orders created in Magento from Odoo?

A: Orders are created via the Magento Guest Cart API. The connector creates a cart, adds items by SKU, sets addresses and payment/shipping methods, then places the order.

Q: What happens when I validate a delivery in Odoo?

A: The connector automatically creates a shipment in Magento with the corresponding item quantities.

Q: What happens when I post an invoice in Odoo?

A: The connector automatically creates an invoice in Magento for the linked order.

Q: Can I sync only specific data types?

A: Yes. The Auto Sync tab lets you select which entities to sync (Products, Customers, Categories, Orders).

Q: How does bidirectional field mapping work?

A: Outbound: when pushing to Magento, mapped Odoo field values are injected into the API payload. Inbound: when pulling from Magento, API response values are set on mapped Odoo fields.

Q: What is the Mapping Report?

A: When you click "Test" on a mapping, it runs against a sample record and generates a report showing what values would change and the full API payload that would be sent.

Q: How do I handle credit memos/refunds?

A: Credit memos are automatically synced from Magento during order sync. They appear as linked records on the invoice with refund amounts and status.

Q: Does the connector auto-sync on save?

A: Yes. Products, customers, and orders auto-push to Magento when created or updated in Odoo. Use skip_magento_sync context flag to disable this behavior.

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For support and updates, contact the SDLC Corp team.

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